

CHAP.16

FLUID DYNAMICS

Fluid Dynamics

Two ways of describing the motion of a fluid:

One is developed by Lagrange (Follow particles):

Solution for the positions and velocities of each particle as function of the time

Another is developed by Euler (Distribution Function):

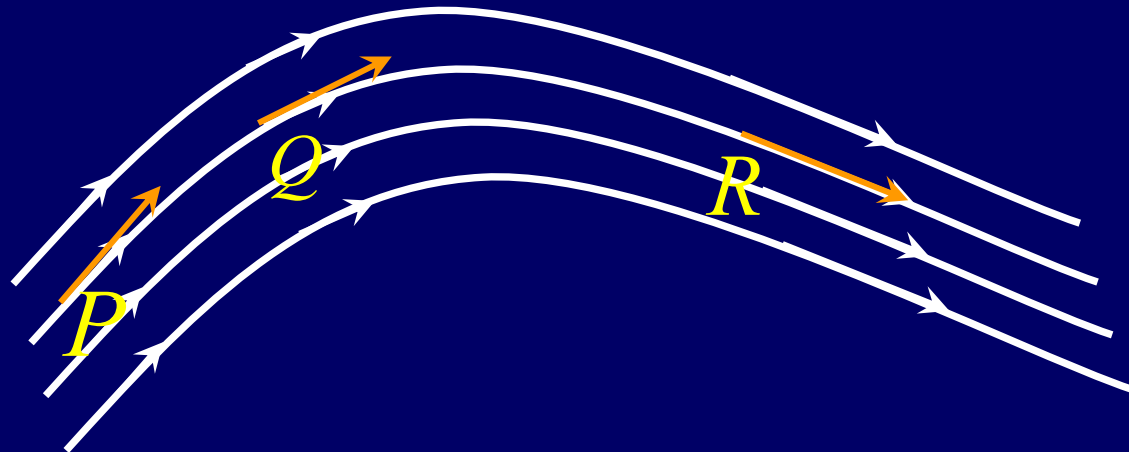
Specify the density and the velocity of the fluid at each point in space and at each instant of time

❖ Description of Fluid Flow:

— Variables: **Pressure, Density, Velocity**

Pressure $p(x,y,z,t)$, **density** $\rho(x,y,z,t)$ and **velocity** $v(x,y,z,t)$ at the point x,y,z in space at a particular time.

Fluid flow



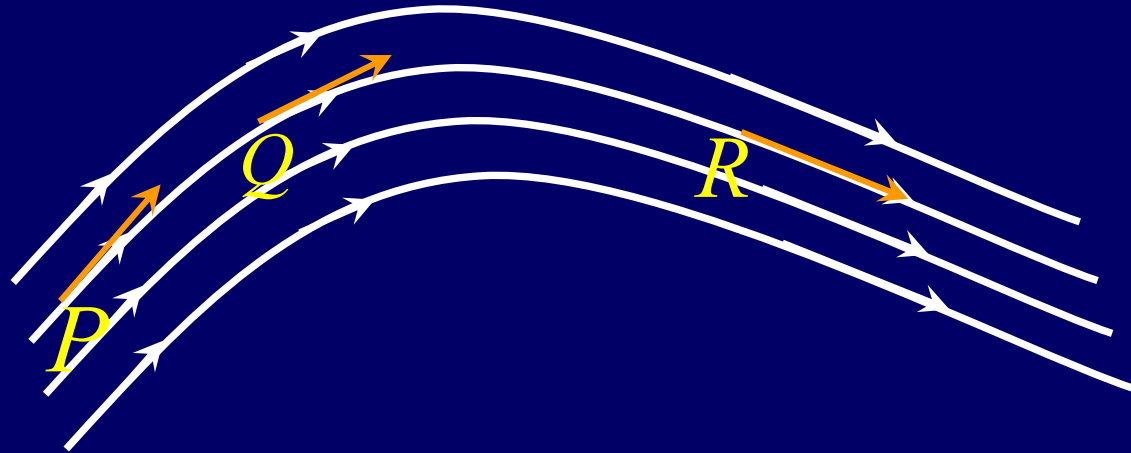
❖ Characteristics of Fluid Flow:

- Steady or not: Variables change with time?
- Compressible or not
- Viscous or not
- Rotational or irrotational

Ideal fluid is steady, incompressible, nonviscous and irrotational fluid.

Streamlines and the Equation of Continuity

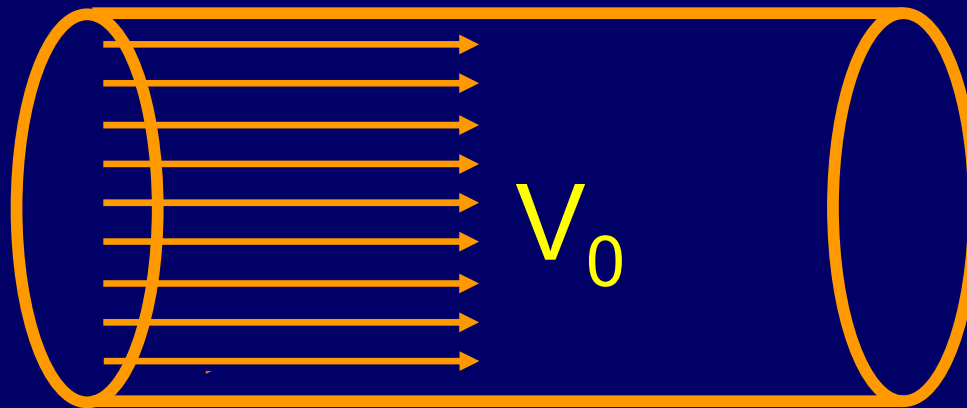
Streamlines:



Fluid flow

viscous or nonviscous

*Kinetic energy can be
transformed into internal energy*

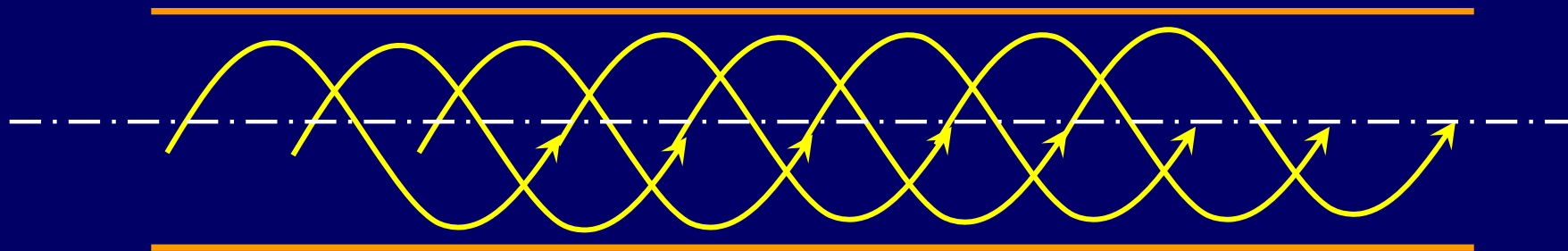


Fluid flow

Rotational or irrotational

rotate about an axis

though its center of mass

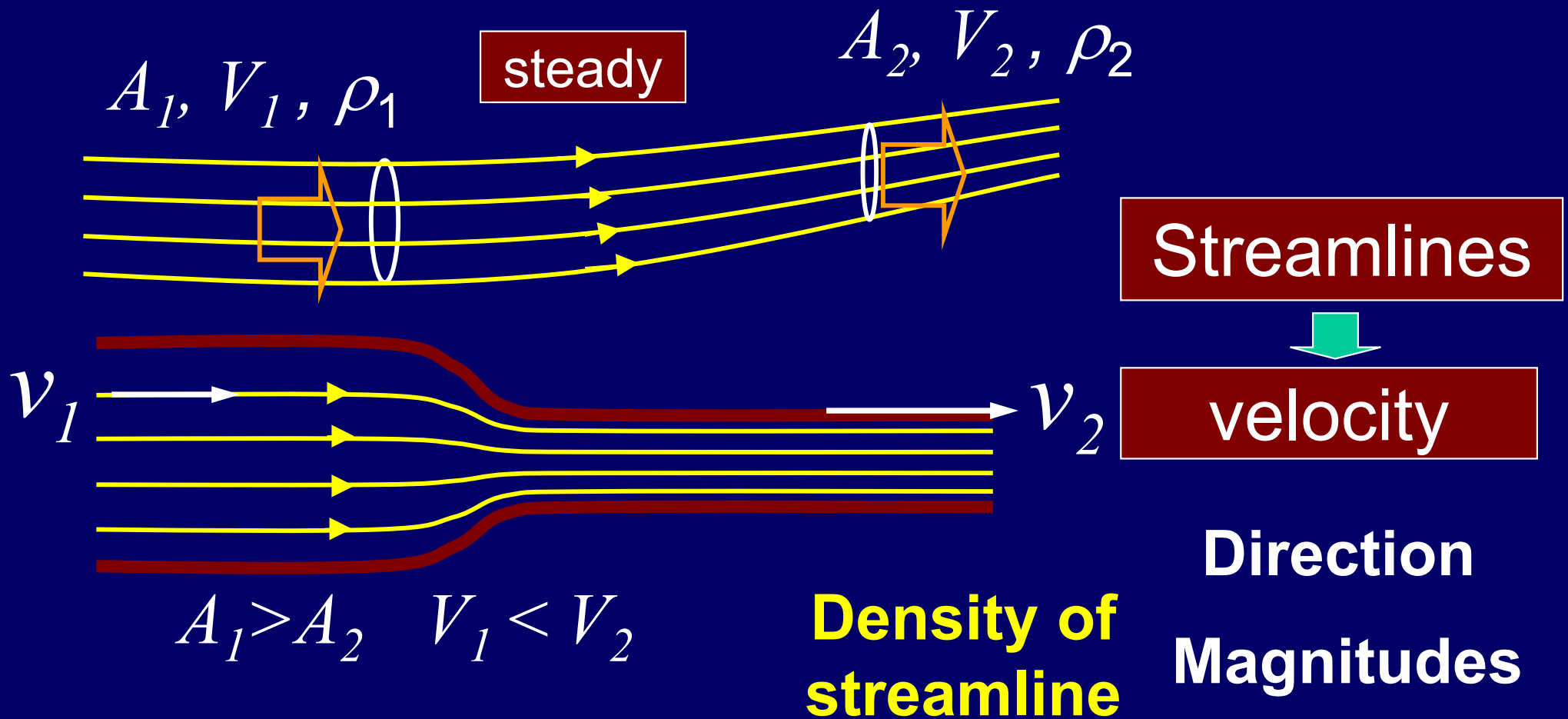


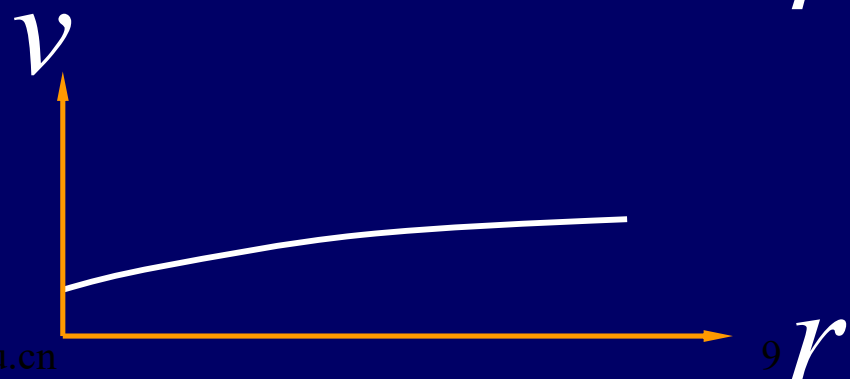
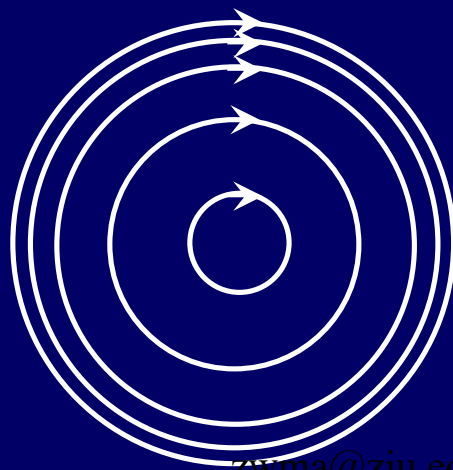
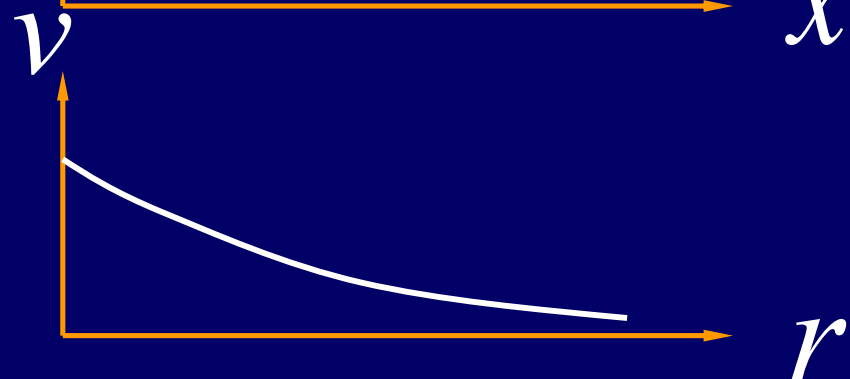
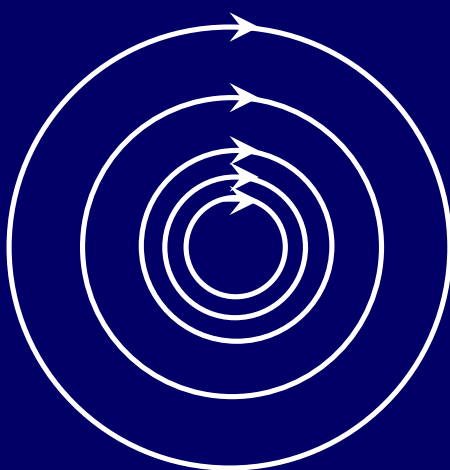
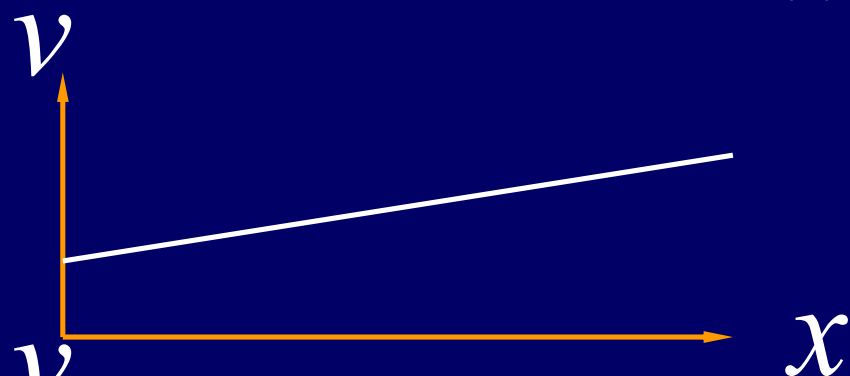
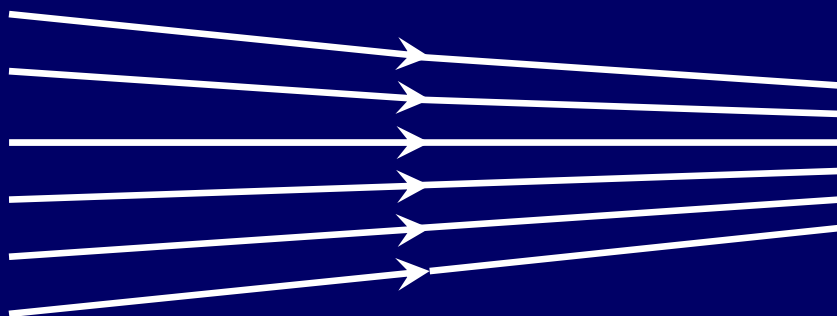
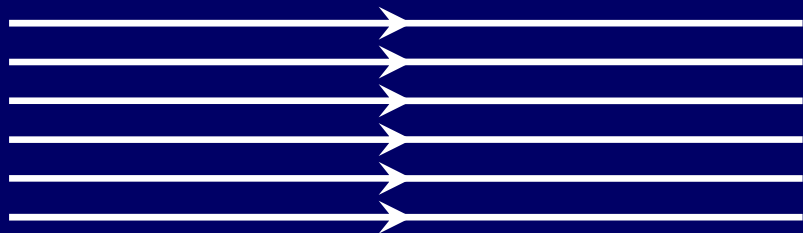
Fluid flow

Equation of continuity

$$A_1 V_1 \rho_1 = A_2 V_2 \rho_2 \quad (\text{No fluid can cross the boundary of flow tube})$$

$$A_1 V_1 = A_2 V_2 \quad \text{If } \rho_1 = \rho_2 \text{ (incompressible)}$$





$$p_1 A_1 v_1 - p_2 A_2 v_2 = \frac{1}{2} A_2 v_2 \rho v_2^2 - \frac{1}{2} A_1 v_1 \rho v_1^2 + A_2 v_2 \rho g h_2 - A_1 v_1 \rho g h_1$$

$$p_1 - p_2 = \frac{1}{2} \rho v_2^2 - \frac{1}{2} \rho v_1^2 + \rho g h_2 - \rho g h_1$$

$$p_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = p_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

nonviscous

In 1 second

steady

δm

$h_2 A_2 v_2 \rho$

$h_1 A_1 v_1 \rho$

$\leftarrow P_2$

$P_1 \rightarrow$
 δm

Bernoulli's Eq.

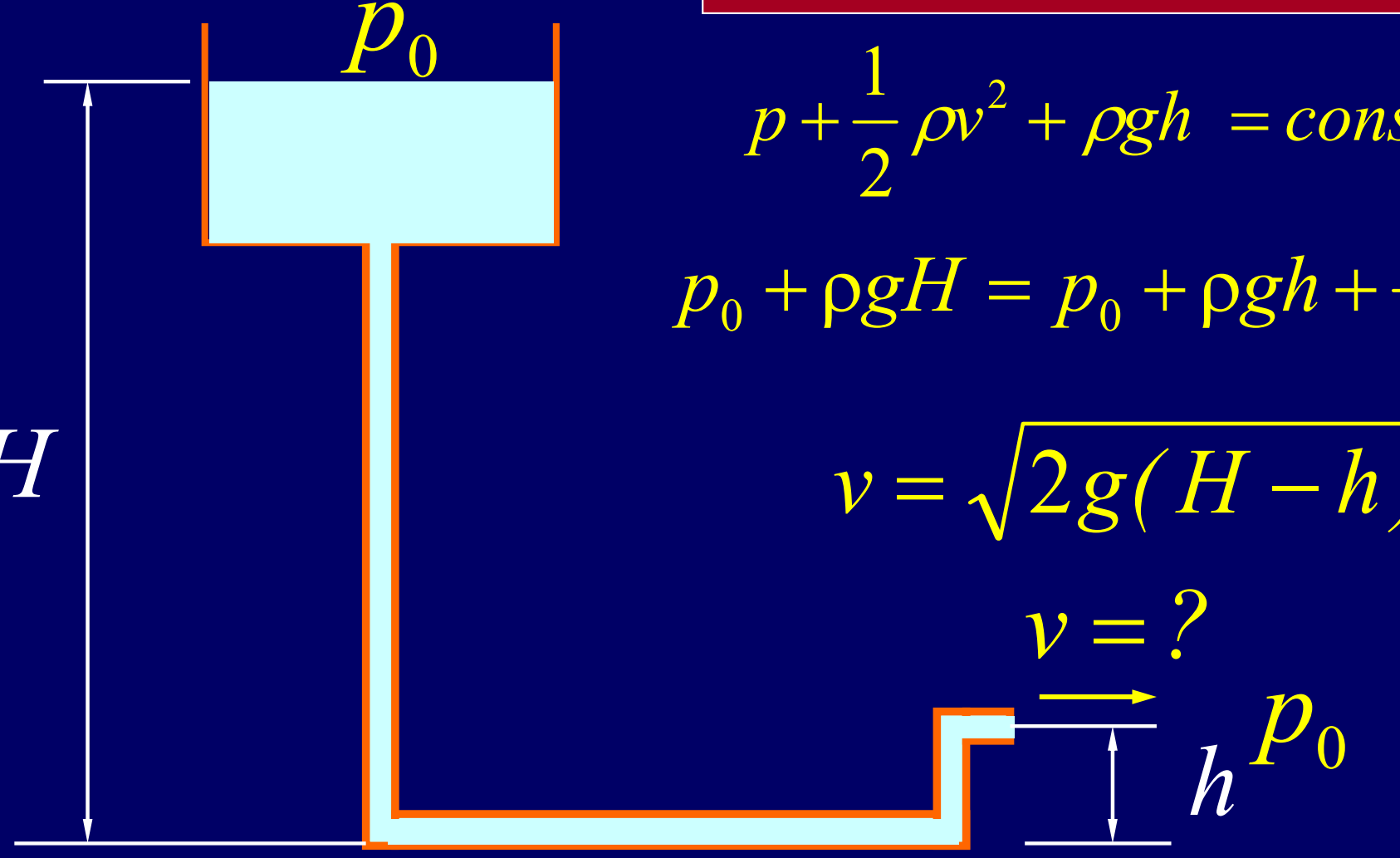
$$p + \frac{1}{2} \rho v^2 + \rho g h = \text{const.}$$

Bernoulli's Eq.

$$p + \frac{1}{2} \rho v^2 + \rho gh = \text{const.}$$

$$p_0 + \rho gH = p_0 + \rho gh + \frac{1}{2} \rho v^2$$

$$v = \sqrt{2g(H - h)}$$

$$v = ?$$


p_0

H

h

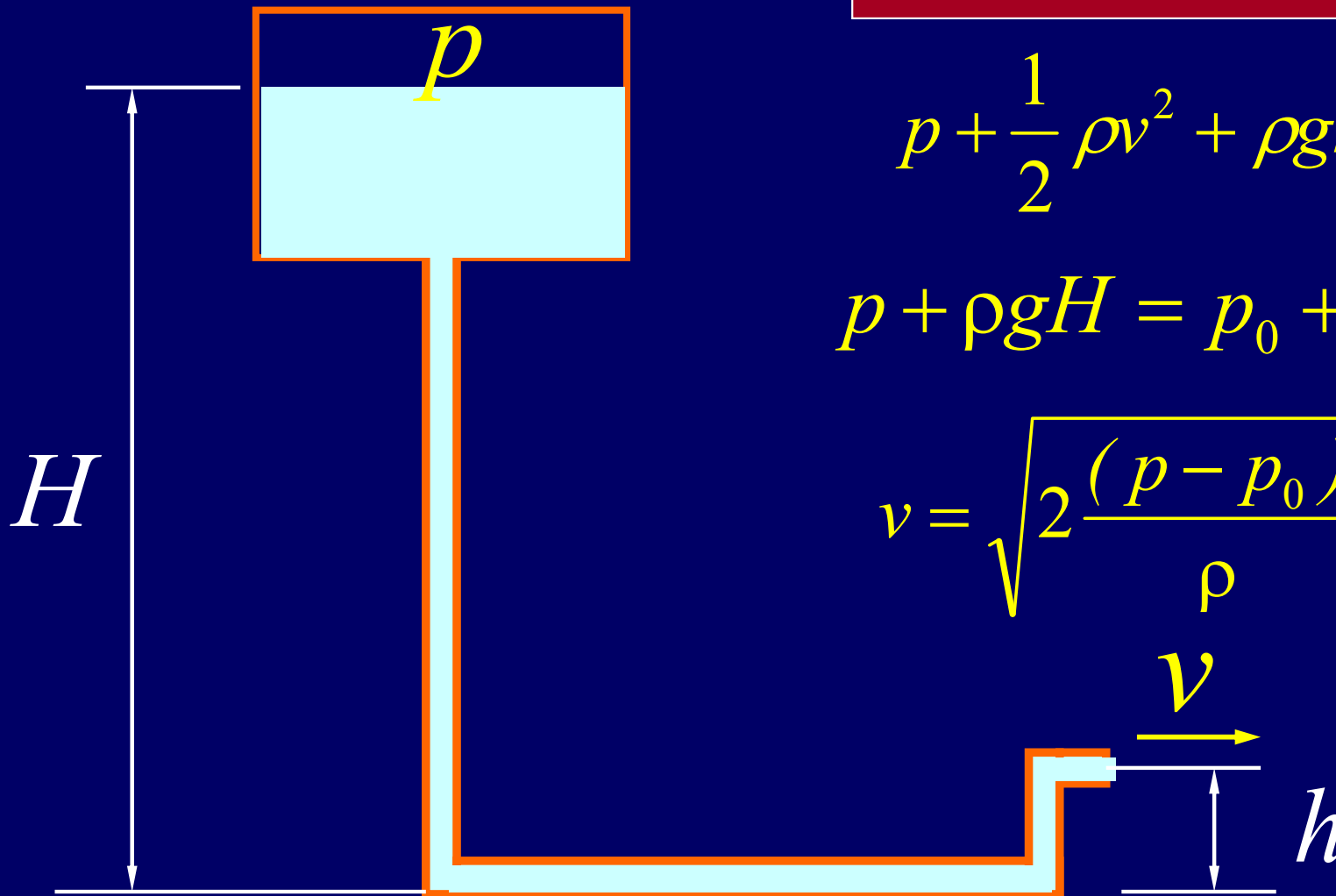
p_0

Bernoulli's Eq.

$$p + \frac{1}{2} \rho v^2 + \rho g h = \text{const.}$$

$$p + \rho g H = p_0 + \rho g h + \frac{1}{2} \rho v^2$$

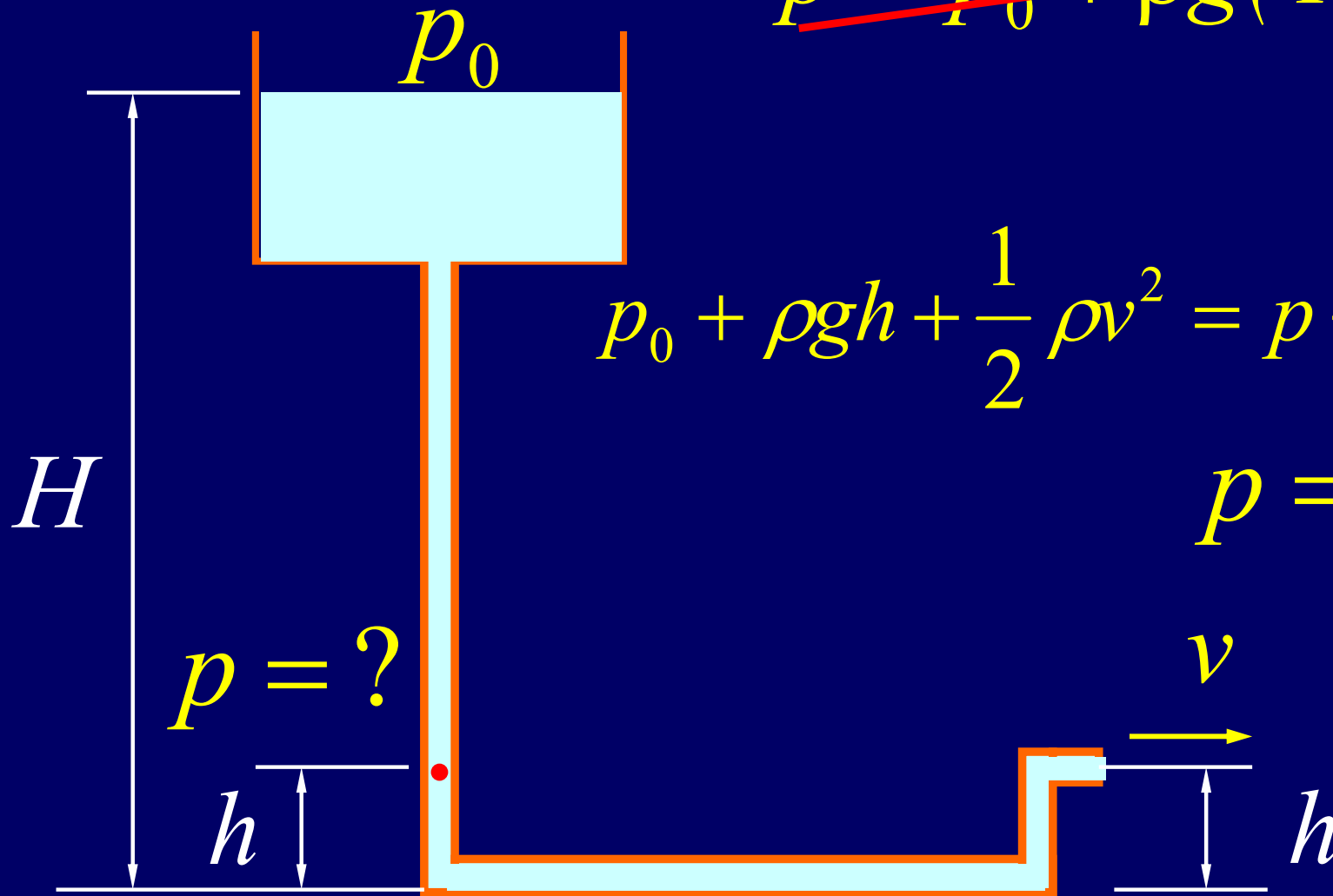
$$v = \sqrt{2 \frac{(p - p_0)}{\rho} + 2g(H - h)}$$

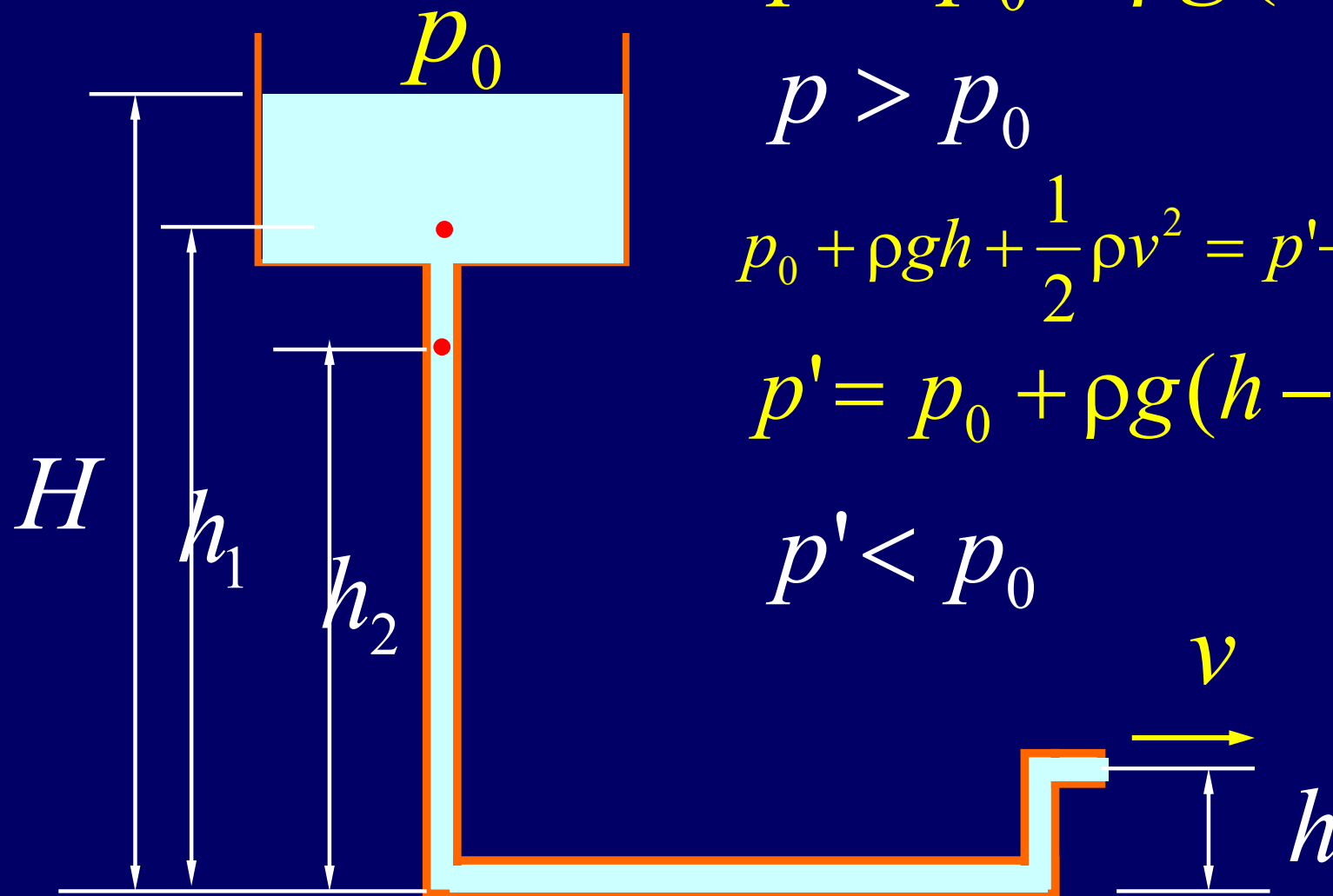


~~$$p = p_0 + \rho g(H - h) ???$$~~

$$p_0 + \rho gh + \frac{1}{2} \rho v^2 = p + \rho gh + \frac{1}{2} \rho v^2$$

$$p = p_0$$





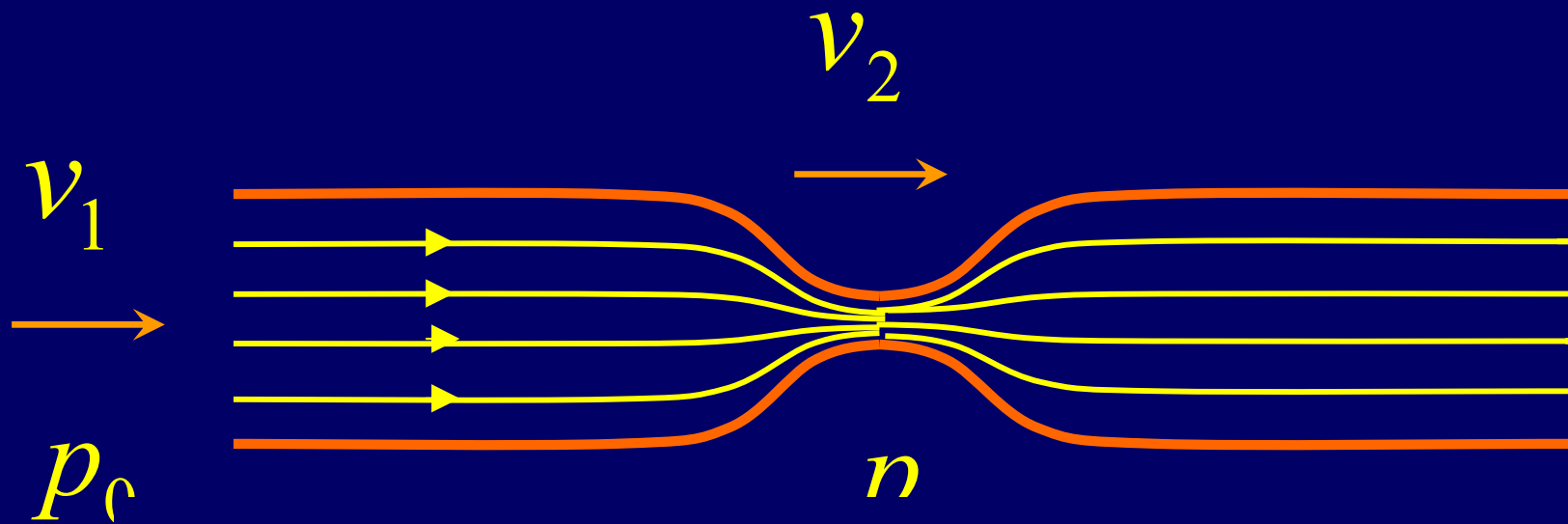
$$p = p_0 + \rho g(H - h_1)$$

$$p > p_0$$

$$p_0 + \rho g h + \frac{1}{2} \rho v^2 = p' + \rho g h_2 + \frac{1}{2} \rho v^2$$

$$p' = p_0 + \rho g(h - h_2)$$

$$p' < p_0$$



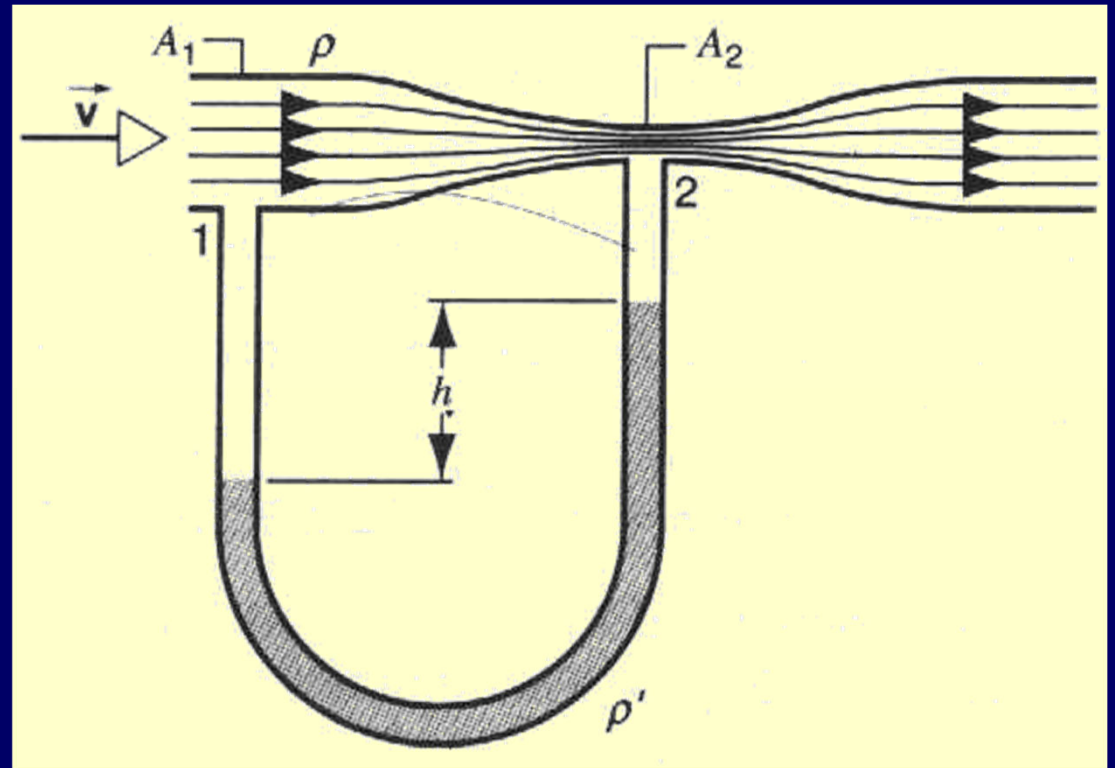
$$p_0 + \rho gh + \frac{1}{2} \rho v_1^2 = p + \rho gh + \frac{1}{2} \rho v_2^2$$

$$p_0 + \frac{1}{2} \rho v_1^2 = p + \frac{1}{2} \rho v_2^2 \quad v_1 < v_2 \quad p_0 > p$$

Applications of Bernoulli's Equation and the Equation of Continuity

1. The Venturi meter

$$v_1 = A_2 \sqrt{\frac{2(\rho' - \rho)gh}{\rho(A_1^2 - A_2^2)}}$$



How to get this result?

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

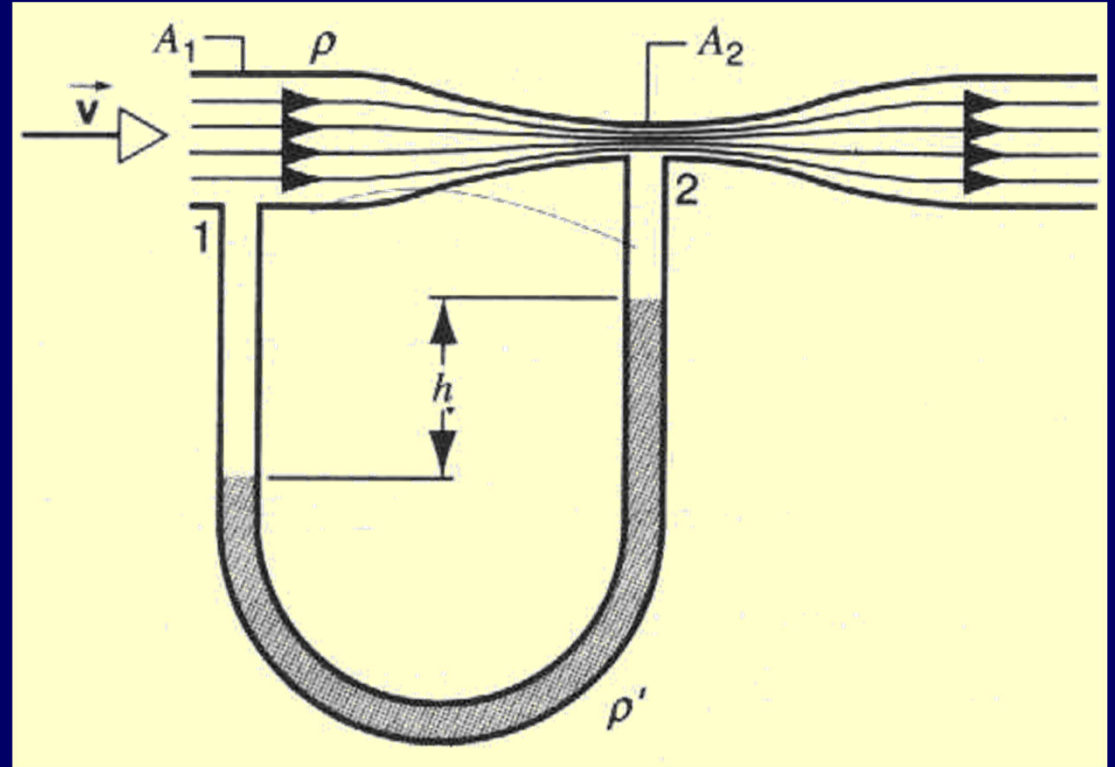
$$p_1 - p_2 = \frac{1}{2}\rho(v_2^2 - v_1^2)$$

Mass conservation:

$$v_1 A_1 = v_2 A_2$$

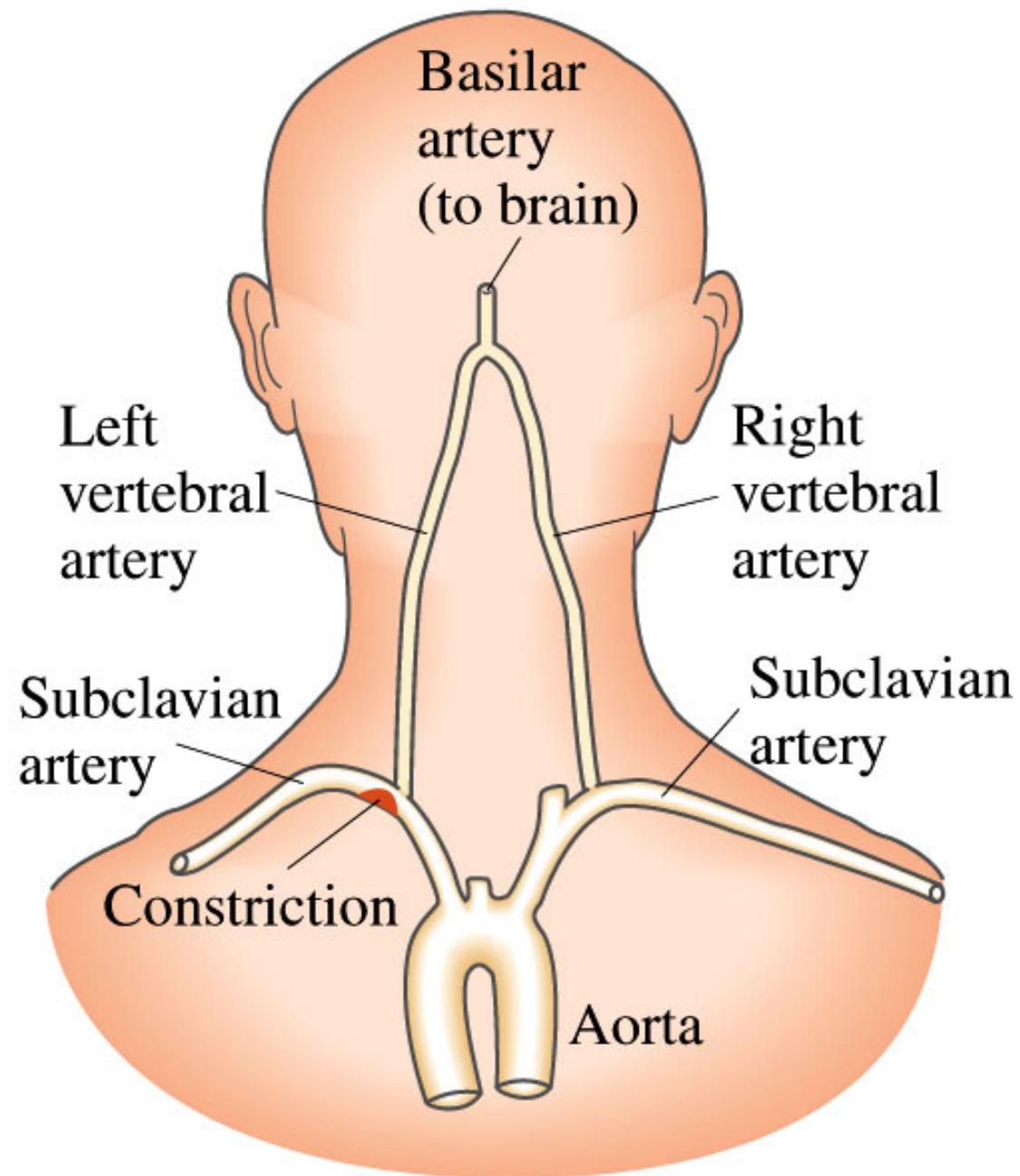
$$p_1 - p_2 = \frac{1}{2}\rho\left(\frac{A_1^2}{A_2^2}v_1^2 - v_1^2\right)$$

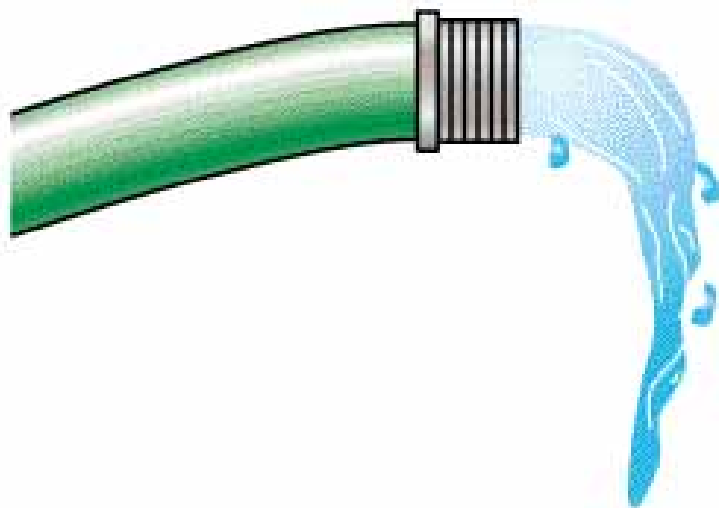
$$= \frac{v_1^2}{2A_2^2}\rho(A_1^2 - A_2^2)$$



$$p_1 - p_2 = (\rho' - \rho)gh$$

$$v_1 = A_2 \sqrt{\frac{2(\rho' - \rho)gh}{\rho(A_1^2 - A_2^2)}}$$



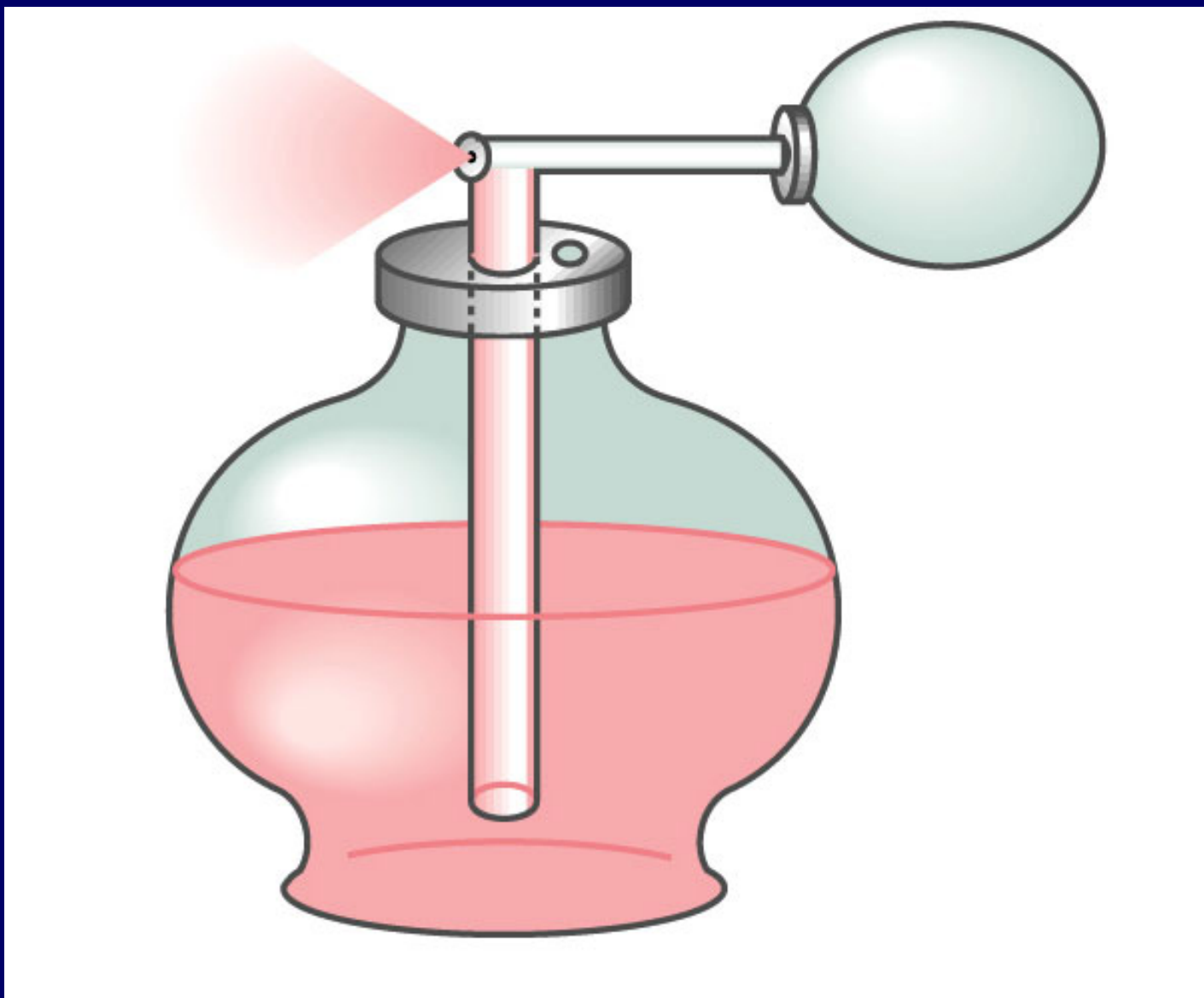


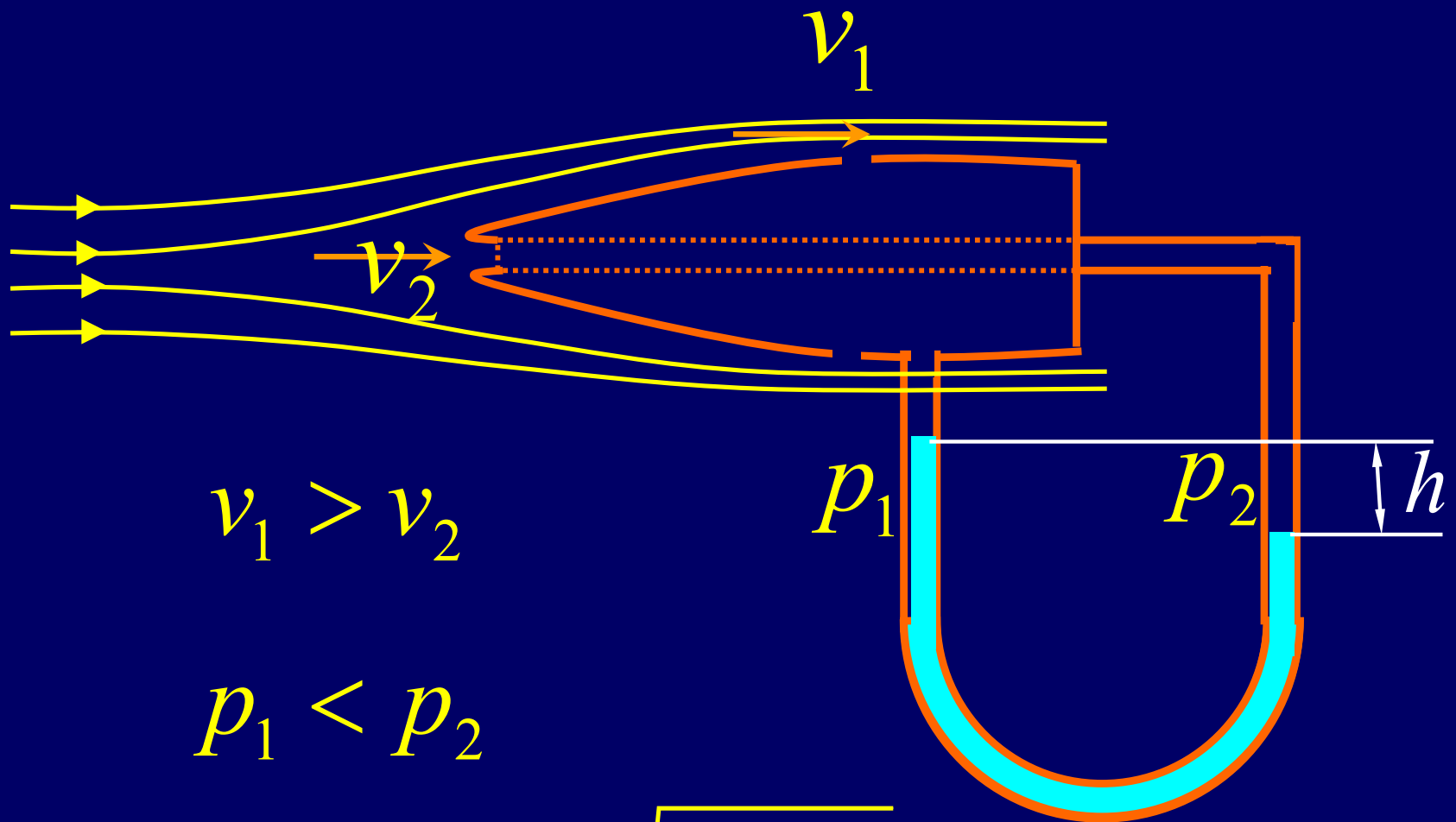


Geiser



© 2007 Thomson Higher Education

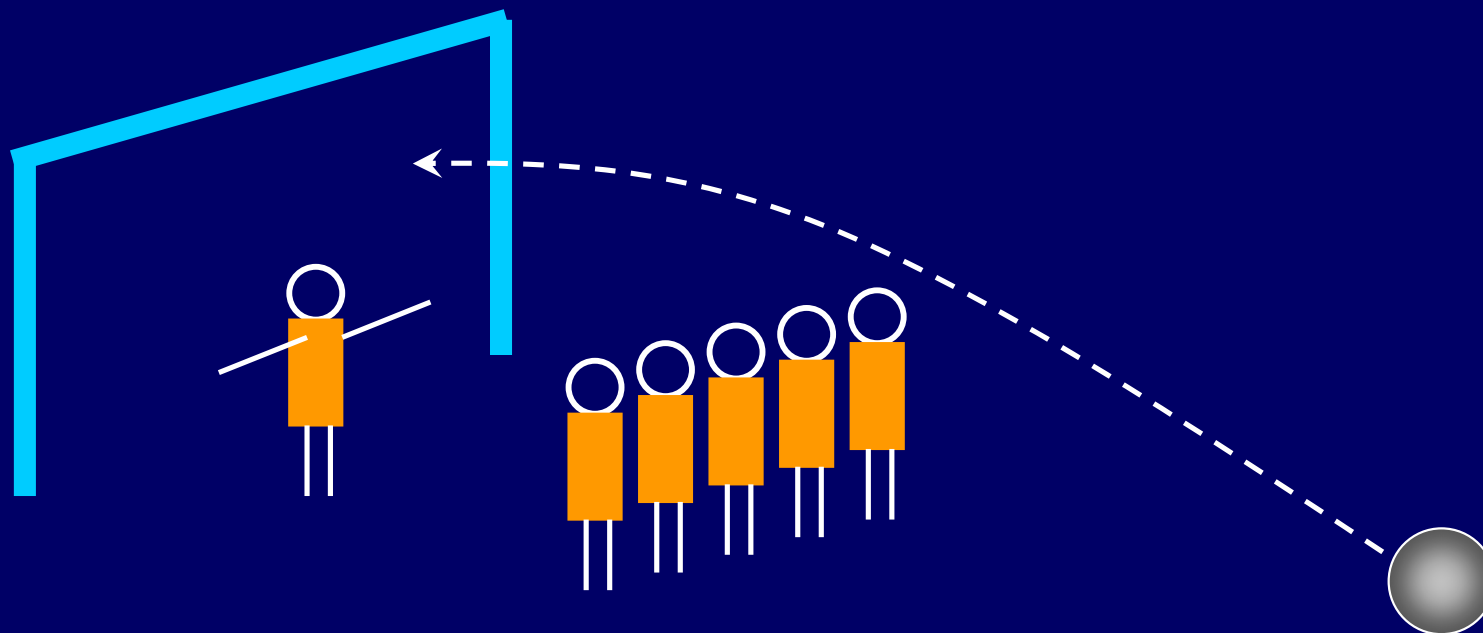


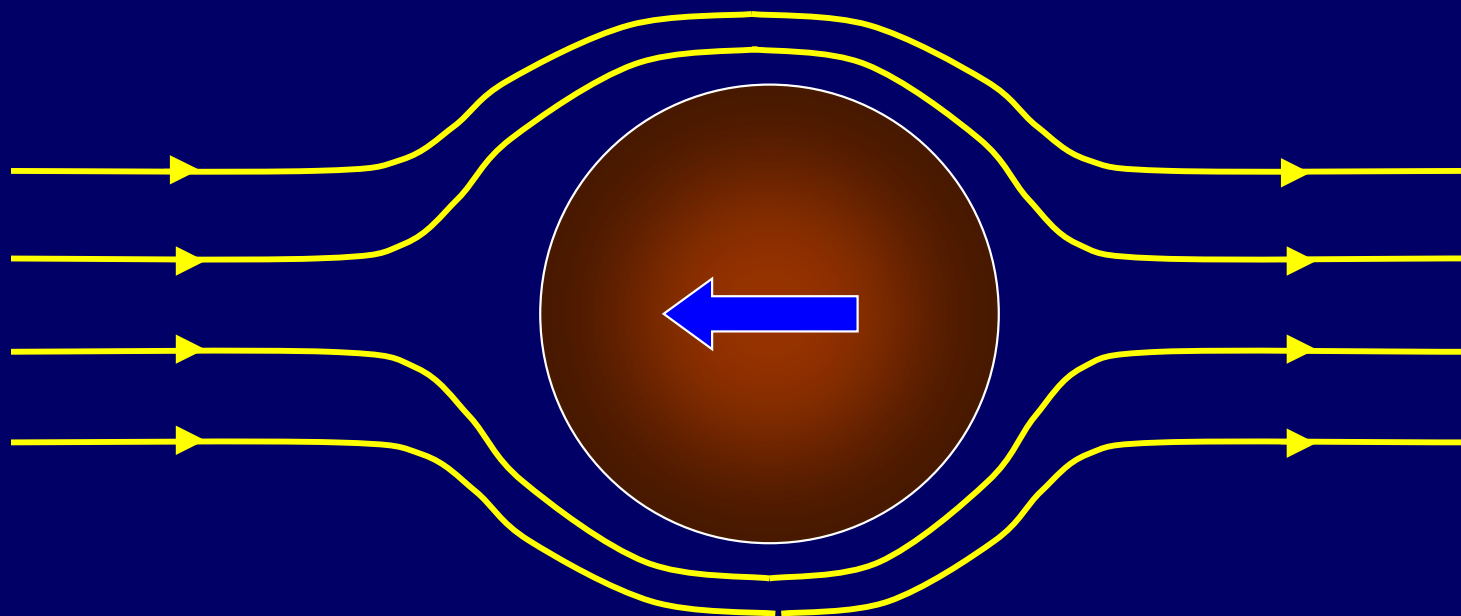


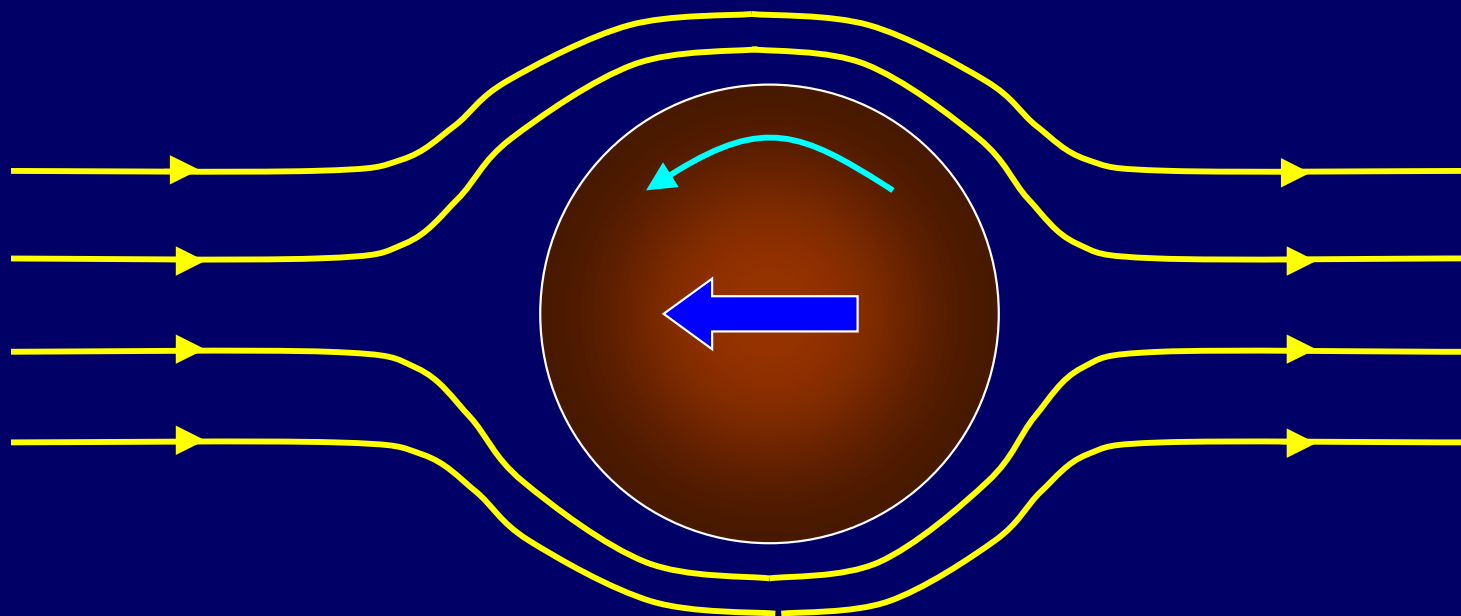
$$v_1 > v_2$$

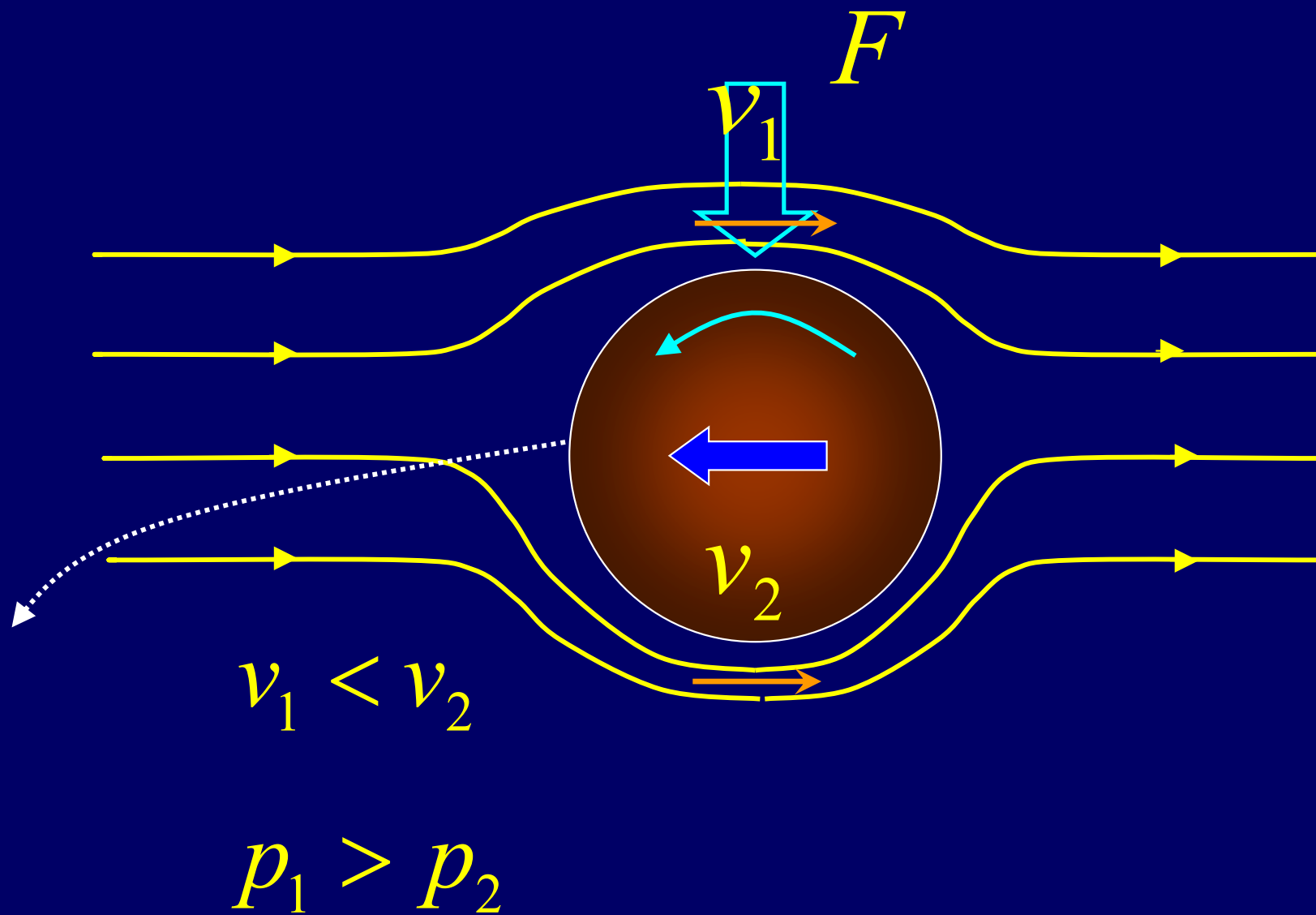
$$p_1 < p_2$$

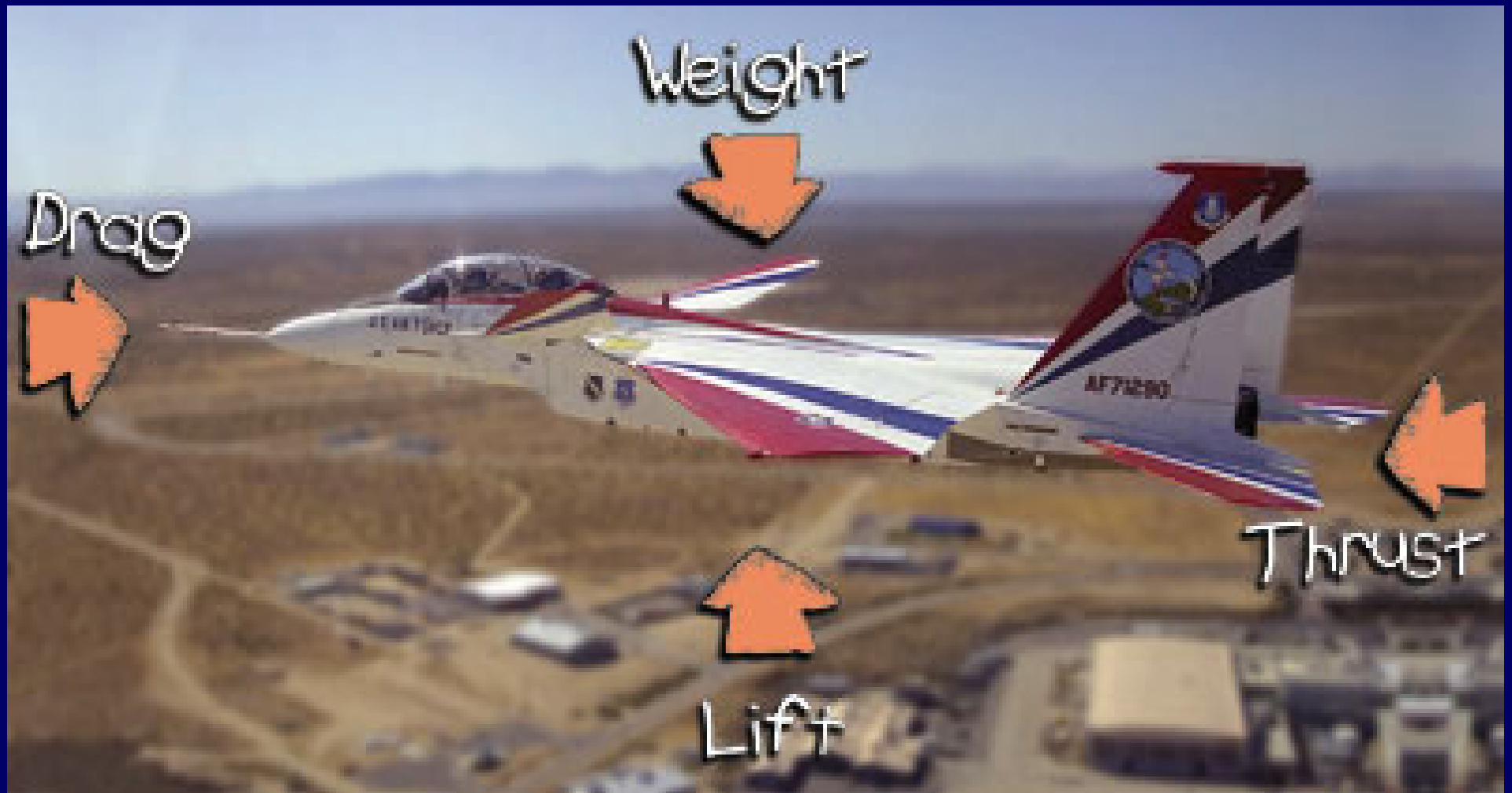
$$v_1 = \sqrt{\frac{2gh\rho'}{\rho}}$$

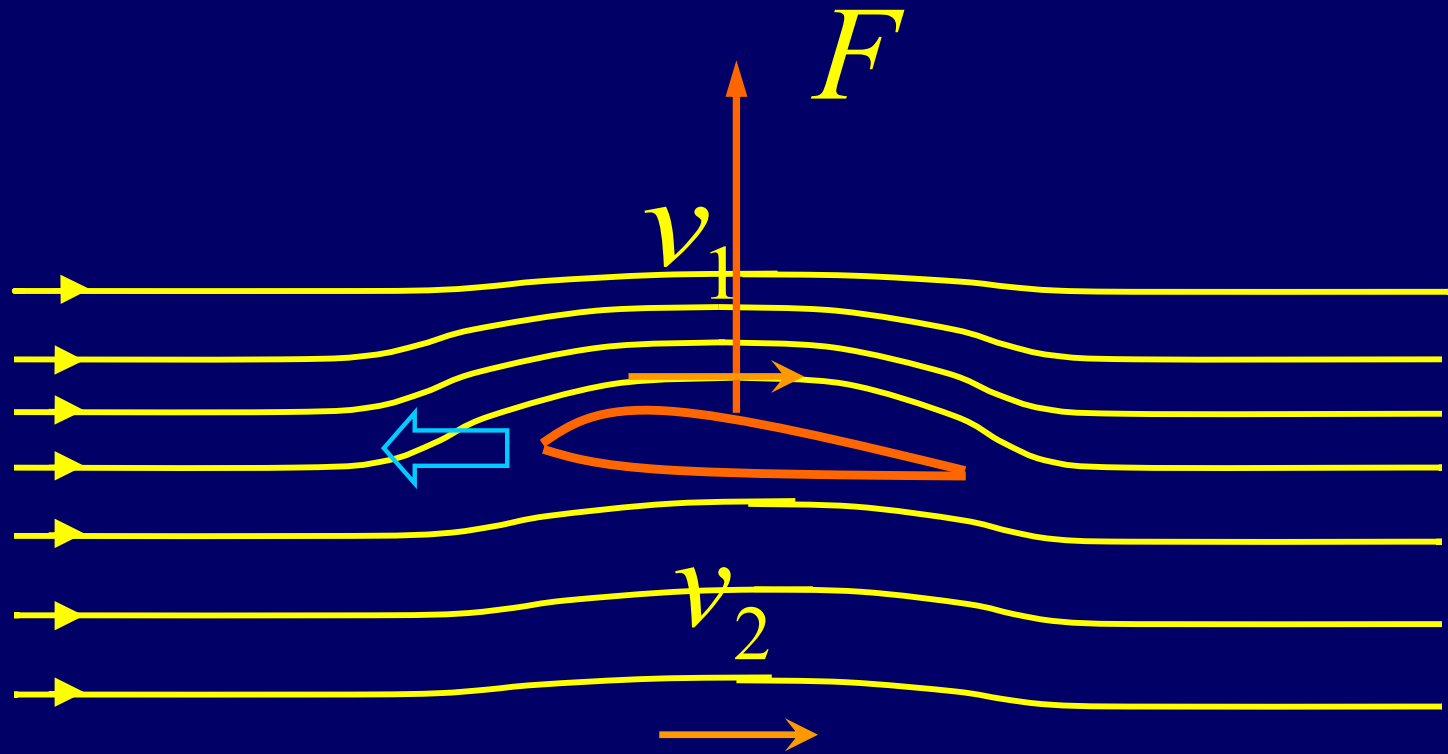












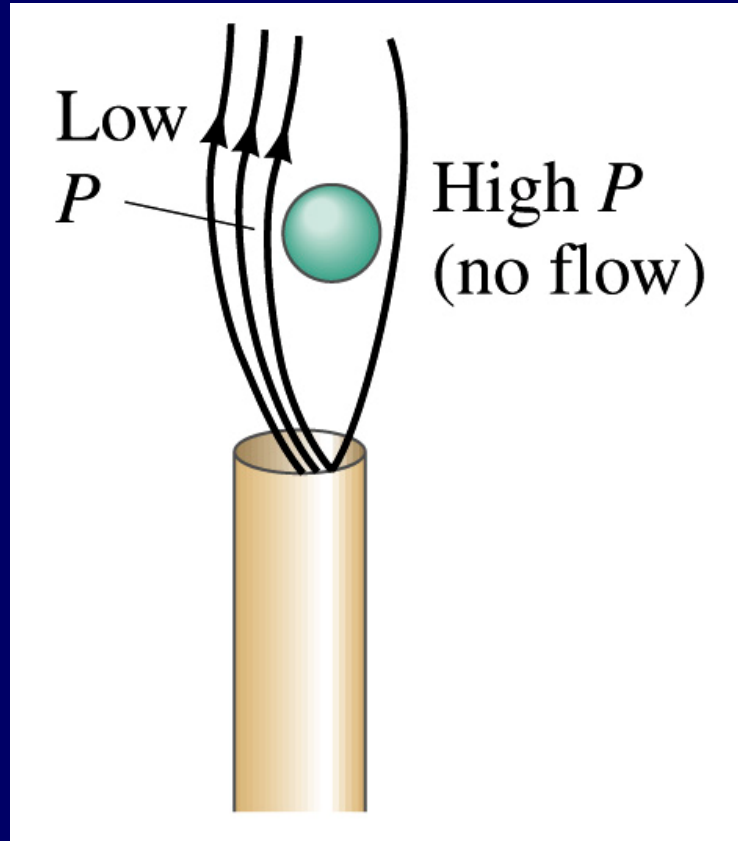
$$v_1 > v_2$$

$$p_1 < p_2$$

The wing of airplane

Now, can you explain these phenomena ?





Rocket



$$p_0 + \frac{1}{2} \rho v_0^2 = p$$

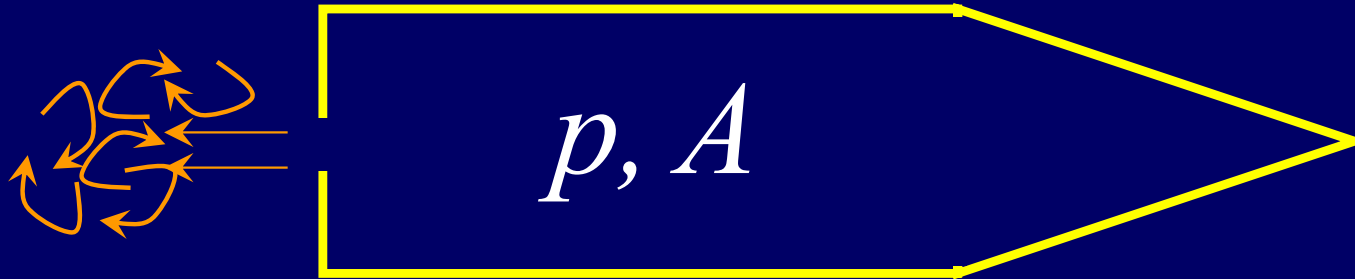
$$\frac{dm}{dt} = \frac{\rho A_0 v_0^2}{v_0}$$

$$v_0 = \sqrt{\frac{2(p - p_0)}{\rho}}$$

$$\frac{dm}{dt} = 2 \frac{A_0}{v_0} (p - p_0)$$

$$F = \frac{dp}{dt} = \frac{dm v_0}{dt} = 2 A_0 (p - p_0)$$

Rocket

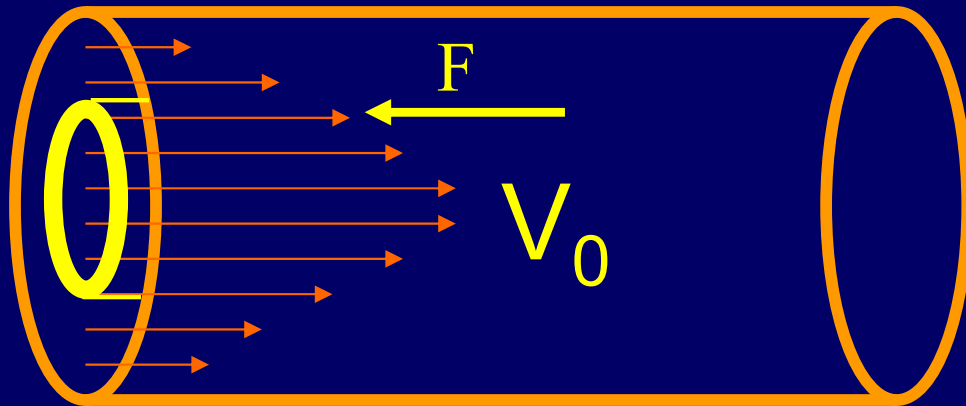
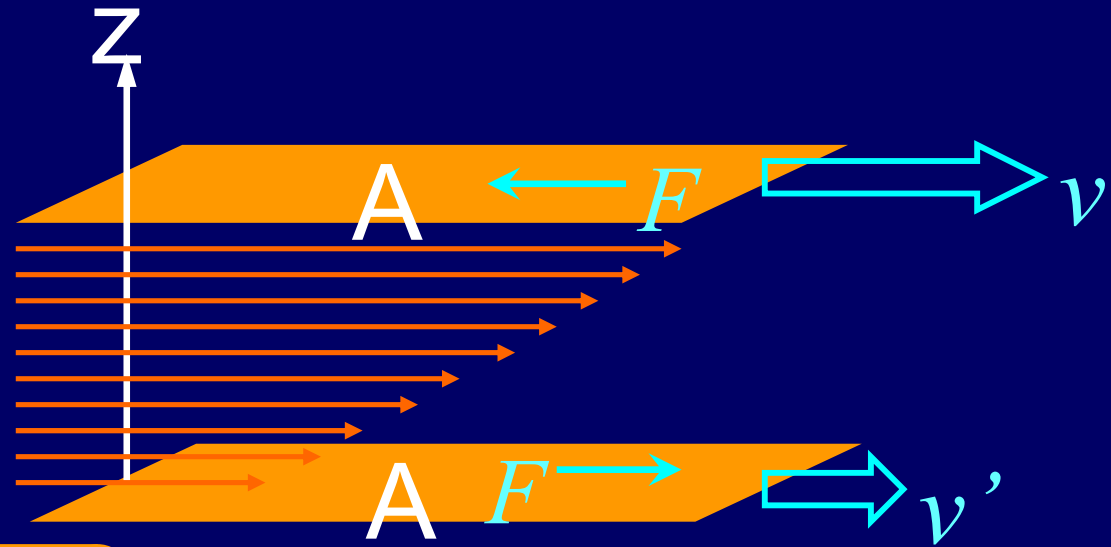


Turbulence

Bernoulli's Eq. cannot be used

viscous

$$F = A \eta \, dv/dZ$$



$$F = 2\pi r L \eta \, dv/dr$$

$$F_{ex} = S(p_2 - p_1) = \pi r^2 \Delta p$$

Since $F_{ex} = F$, we have

$$dv = (\Delta p / 2L \eta) r \, dr$$

$$v = v_0 (1 - r^2/R^2)$$

$$\text{where } v_0 = \Delta p R^2 / 4L \eta$$

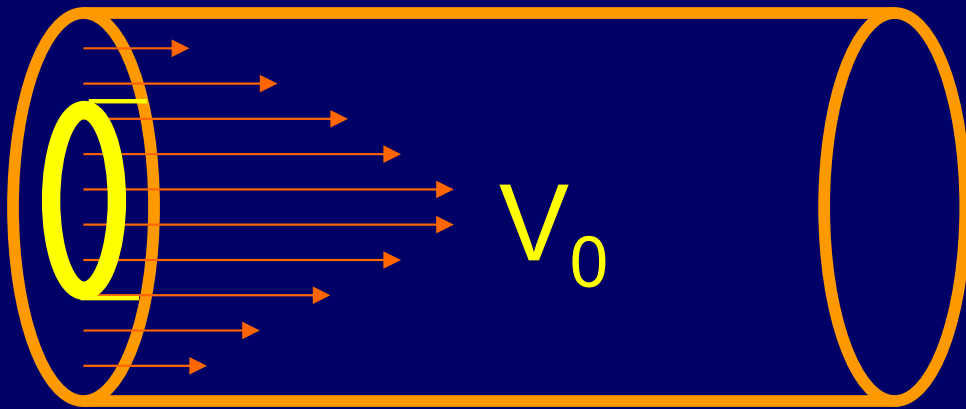
$$dm = 2\pi r dr v dt \rho$$

$$\frac{dm}{dt} = \int_0^R 2\pi r \rho v_0 \left(1 - \frac{r^2}{R^2}\right) dr$$

$$= \frac{1}{2} \pi \rho v_0 R^2$$

$$= \frac{1}{2} \pi \rho \left(\frac{\Delta p R^2}{4L\eta} \right) R^2$$

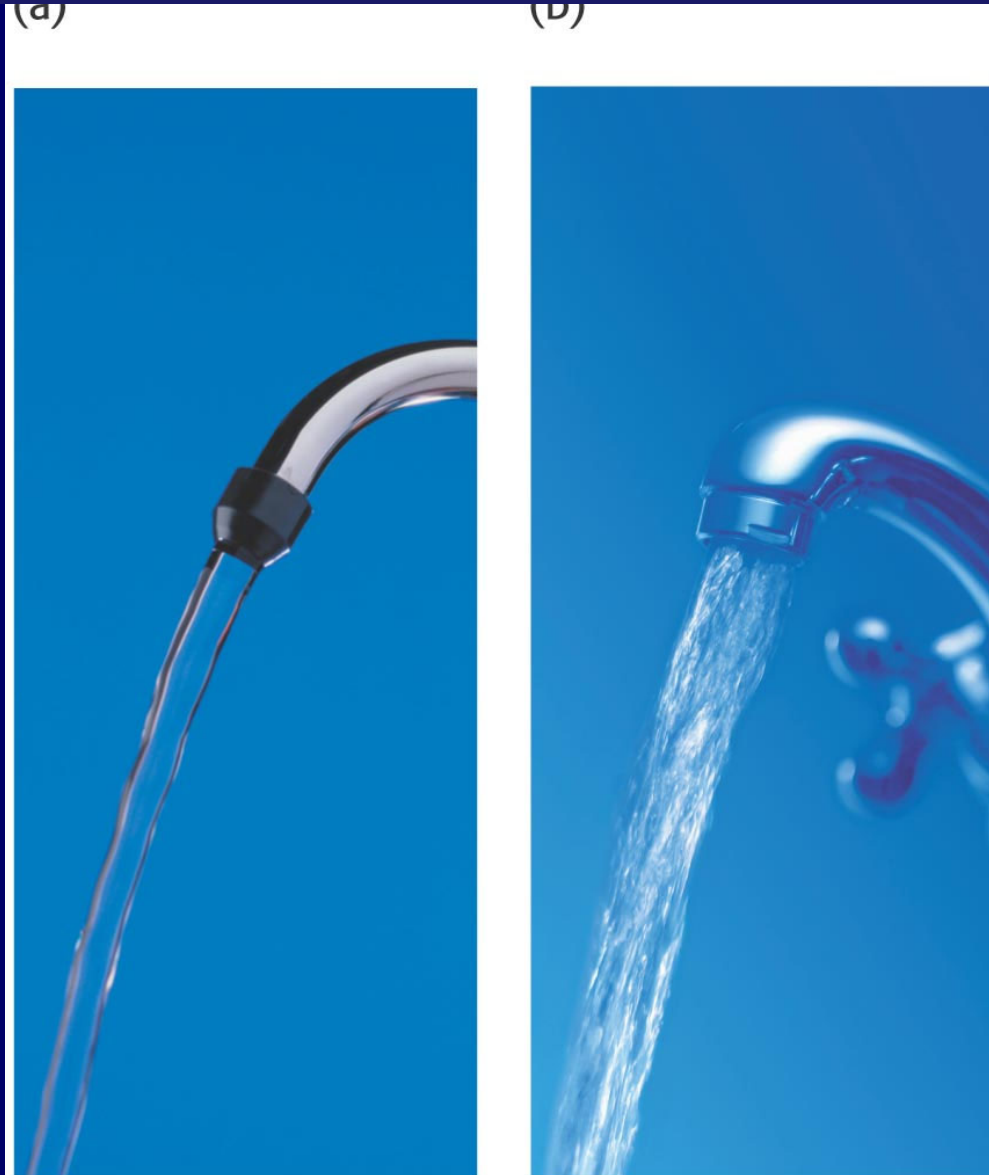
$$\frac{dm}{dt} = \frac{\pi \rho \Delta p R^4}{8L\eta}$$



$$v = v_0 \left(1 - r^2/R^2\right)$$

Turbulence:

Turbulent (disordered) flow — Chaotic flow



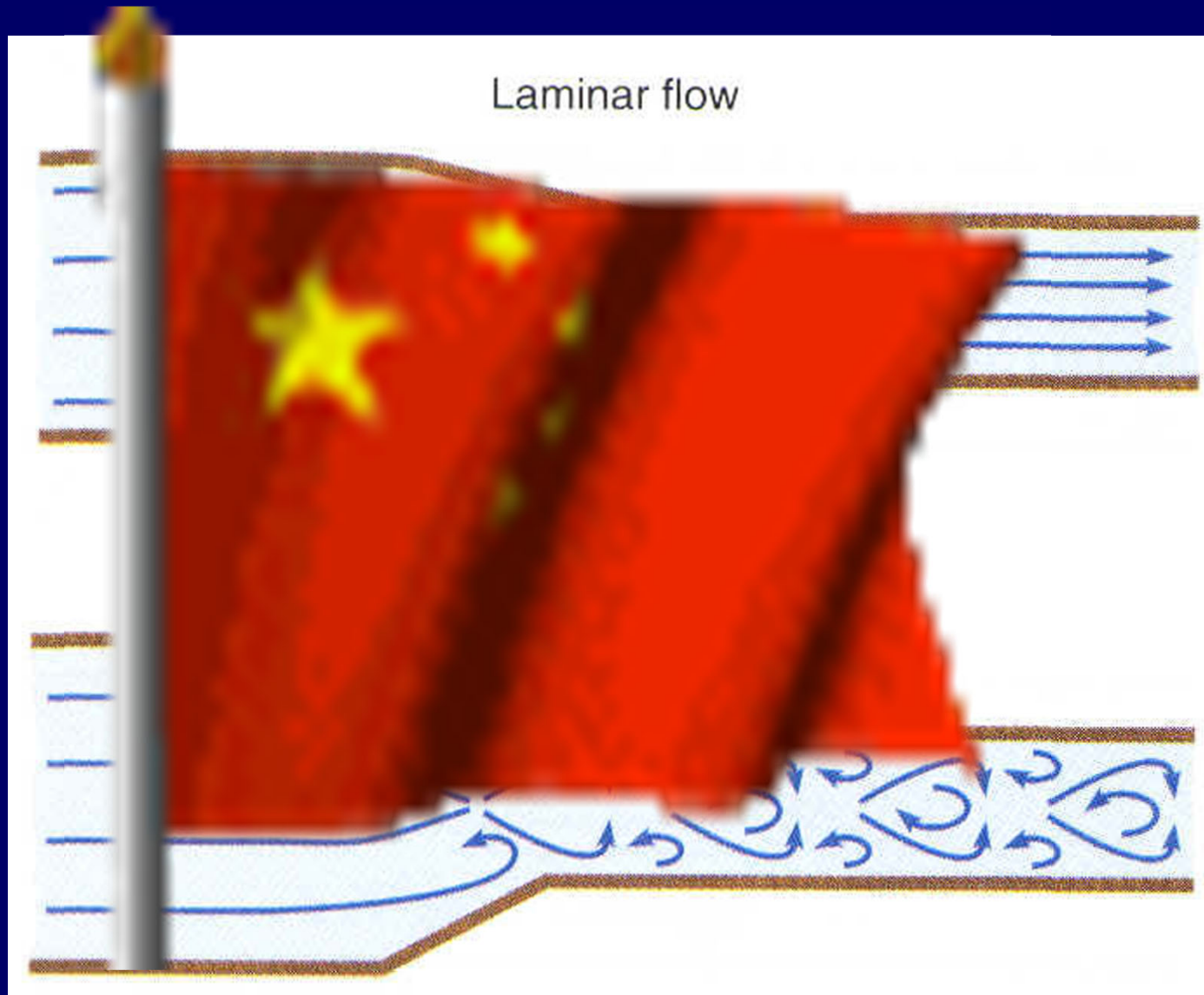
Copyright © 2008 Pearson Education, Inc., publishing as Pearson Addison-Wesley.



© 2007 Thomson Higher Education



Copyright © 2008 Pearson Education, Inc., publishing as Pearson Addison-Wesley.



Turbulence

$$v_c \propto \eta^a \rho^b D^c \quad LT^{-1} = (ML^{-1}T^{-1})^a (ML^{-3})^b (L)^c$$

$$a = 1, b = -1, c = -1$$

Critical speed:
$$v_c = R \frac{\eta}{\rho D}$$

R is Reynolds number, D is diameter of a tube

The critical speed for water in the pipe with $D \sim 2\text{cm}$

for our home, $R \sim 2000$, $D = 0.02\text{m}$, $\eta = 1 \times 10^{-3} \text{ N} \cdot \text{s} / \text{m}^2$

$$v_c \approx 10 \text{ cm} / \text{s}$$

CHAP.16

Exercises

P367: 7,10,17, 21,23

Problems

P369: 4, 11, 14